

Numpy

```
In [3]: import numpy as np
```

```
In [5]: np.__version__
```

```
Out[5]: '2.3.5'
```

```
In [2]: list=[1,2,3,4,5,6]  
list
```

```
Out[2]: [1, 2, 3, 4, 5, 6]
```

```
In [8]: arr=np.array(list)  
arr
```

```
Out[8]: array([1, 2, 3, 4, 5, 6])
```

```
In [10]: print(type(list))  
print(type(arr))  
  
<class 'list'>  
<class 'numpy.ndarray'>
```

arange

```
In [11]: np.arange(4)      # arange work as range
```

```
Out[11]: array([0, 1, 2, 3])
```

```
In [12]: np.arange(1,10)
```

```
Out[12]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [14]: np.arange(1,10,2)
```

```
Out[14]: array([1, 3, 5, 7, 9])
```

```
In [16]: np.arange(10,20,3,4)  # 3 arg allow
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[16], line 1  
----> 1 np.arange(10,20,3,4)    # 3 arg allow  
  
TypeError: Cannot interpret '4' as a data type
```

```
In [17]: np.arange(20,10)      # 1st arg should be always small
```

```

-----
NameError                                Traceback (most recent call last)
Cell In[17], line 1
----> 1 np.arange(20,10)

NameError: name 'np.arange' is not defined

```

```
In [19]: np.arange(-20,10)
```

```
Out[19]: array([-20, -19, -18, -17, -16, -15, -14, -13, -12, -11, -10, -9, -8,
              -7, -6, -5, -4, -3, -2, -1,  0,  1,  2,  3,  4,  5,
              6,  7,  8,  9])
```

```
In [22]: # zeros - bydefault in float
```

```
In [23]: np.zeros(4) # 1d array
```

```
Out[23]: array([0., 0., 0., 0.])
```

```
In [25]: np.zeros([2,2]) # 2d array
```

```
Out[25]: array([[0., 0.],
               [0., 0.]])
```

```
In [28]: np.zeros([2,2],dtype=int)
```

```
Out[28]: array([[0, 0],
               [0, 0]])
```

```
In [29]: np.zeros([3,4])
```

```
Out[29]: array([[0., 0., 0., 0.],
               [0., 0., 0., 0.],
               [0., 0., 0., 0.]])
```

```
In [30]: np.zeros([10,10],dtype=int)
```

```
Out[30]: array([[0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0],
               [0, 0, 0, 0, 0, 0, 0, 0, 0, 0]])
```

ones - bydefault in float

```
In [34]: np.ones([5])
```

```
Out[34]: array([1., 1., 1., 1., 1.])
```

```
In [36]: np.ones([1,10],dtype=int)
```

Out[36]: array([[1, 1, 1, 1, 1, 1, 1, 1, 1, 1]])

In [37]: `np.two(5)` *# their is fun name such as two*

```
-----
AttributeError                                Traceback (most recent call last)
Cell In[37], line 1
----> 1 np.two(5)          # their is fun name such as two

File C:\ProgramData\anaconda3\Lib\site-packages\numpy\__init__.py:795, in __getattr__(attr)
    792     import numpy.char as char
    793     return char.chararray
--> 795 raise AttributeError(f"module {__name__!r} has no attribute {attr!r}")

AttributeError: module 'numpy' has no attribute 'two'
```

random

In [38]: `np.random.rand(5)` *# it gives any random values 5 times*

Out[38]: array([0.29803454, 0.86990354, 0.96451306, 0.36980474, 0.39194368])

In [42]: `np.random.rand(2,5)` *# 2*5 matrix*

Out[42]: array([[0.40503659, 0.28784049, 0.8574148 , 0.70094152, 0.8967477],
 [0.78552245, 0.90242915, 0.45465533, 0.05860523, 0.89520982]])

In [45]: `np.random.rand(1,4,2)` *# 1 block , 4 rows , 2 column*

Out[45]: array([[0.17381968, 0.55685514],
 [0.80586716, 0.94250399],
 [0.15352992, 0.24749226],
 [0.7150395 , 0.6101928]])

In [46]: `np.random.randint(6)` *# randon value 0 to 5*

Out[46]: 3

In [47]: `np.random.randint(10,20)` *# 10 to 19*

Out[47]: 13

In [48]: `np.random.randint(1,10,4)` *# 4 means ,random 4 values*

Out[48]: array([4, 3, 2, 9], dtype=int32)

In [49]: `print(np.random.randint(1,10,4))`

[1 9 9 6]

In [50]: `np.random.randint(1,10,(3,4))` *# 3*4 matrix ,1 to 9*

Out[50]: array([[4, 1, 1, 8],
 [4, 1, 7, 1],
 [9, 1, 9, 5]], dtype=int32)

```
In [51]: np.random.randint(10,20,(2,2))
```

```
Out[51]: array([[15, 19],
               [10, 10]], dtype=int32)
```

```
In [56]: m=np.random.randint(10,30,(3,2))
m
```

```
Out[56]: array([[25, 21],
               [20, 26],
               [19, 24]], dtype=int32)
```

```
In [57]: print(m)      # print = remove , in matrix
```

```
[[25 21]
 [20 26]
 [19 24]]
```

```
In [58]: m[0]      # 0th row
```

```
Out[58]: array([25, 21], dtype=int32)
```

```
In [59]: print(m[0])
```

```
[25 21]
```

```
In [60]: print(m[-2])
```

```
[20 26]
```

```
In [61]: print(m[-1])
```

```
[19 24]
```

```
In [62]: print(m[2,-1])      # 2nd row , -1 column
```

```
24
```

```
In [63]: print(m[-2,-1])
```

```
26
```

```
In [ ]:
```